

7(a)(i)	When two or more waves of the <u>same type meet/superpose</u> {NOT: superimpose} at a point, the <u>resultant displacement</u> {NOT: amplitude} of the waves is equal to the <u>vector sum</u> of their individual displacements at that point.	1 1
(ii) 1.	Two waves are coherent if they have a <u>constant phase difference</u> between them. Do NOT accept: same phase or zero phase difference	1
2.	The two interfering waves coming from a single source are coherent. OR Light from two separate light sources is not coherent.	1
(iii)	Fringe separation, $x = 26 \text{ mm}/12 = \underline{2.17 \text{ mm}}$ $\lambda = ax/D$ $\lambda = \frac{0.38 \times 10^{-3} \times 2.17 \times 10^{-3}}{1.20}$ $= \underline{6.87 \times 10^{-7} \text{ m}}$	A1 M1 A1
(iv)	Light reaching the eye comes from two beams. The <u>direct beam and reflected beam are coherent</u> as they come <u>from the same slit</u>. At locations where the waves are <u>in phase</u>, <u>constructive interference</u> occurs producing <u>maxima intensity</u>. At other locations where the waves are <u>anti-phase</u>, <u>destructive interference</u> occurs producing <u>minima intensity</u>.	1 1
(b)(i)	Where there are <u>depressions</u>, wave <u>has maximum amplitude / is at an antinode</u>. Where there are <u>heaps</u>, <u>wave has zero / minimum amplitude / is at a node</u>.	1 1
(ii)	$2\lambda = 20.0 \text{ cm}$ $\rightarrow \lambda = \frac{0.20}{2} = \underline{0.10 \text{ m}}$ $v = f \lambda$ $f = \frac{330}{0.10} = \underline{3300 \text{ Hz}}$ (ecf_λ)	1 1 1
(iii) 1.	<u>No motion</u> / vibrate with <u>minimum amplitude</u>	1

2.	<u>Vibrate parallel to the length of tube / horizontally with maximum amplitude.</u>	1
(c)(i)	$I_2 = I_1 \cos^2 23^\circ$ $= 147 \cos^2 23^\circ$ $= \underline{125 \text{ W cm}^{-2}}$	1 1
(ii)	Intensity $\propto$ Amplitude², $\therefore \frac{125}{147} = \frac{k A_2^2}{k A_1^2} \Rightarrow \frac{A_2}{A_1} = 0.922$ fractional reduction in amplitude = $\frac{A_1 - A_2}{A_1} = 1 - \frac{A_2}{A_1} = 1 - 0.922 = \underline{0.078}$	1 1